

Prototyp3 Network

Fall 2025 Cohort



Version 1.0.0

“Every great creation starts with a prototype. Welcome to ours.”

19th September, 2025

Welcome to Prototyp3's Fall 2025 Cohort!

A big congrats on officially seizing a spot on our very first Project Cohort. We are so excited to see you bring your creativity and build your skills & experiences over the next 10 weeks.

Your Project Details:

1. **Project Name:** TreatQuest – Train Your Virtual Pet
2. **Provider:** Prototyp3 Network
3. **Project Classification:** Open Challenge ▾
4. **Difficulty Level:** Intermediate ▾
5. **Project Type:** New Project ▾
6. **Area:** Reinforcement Learning, Algorithms, Game Simulation
7. **Time Commitment:** 8-10 hours/week
8. **Primary Platform of Communication:** Discord

Overview

TreatQuest is a playful RL challenge where you teach a virtual pet to collect treats while avoiding “traps” in a 2D grid. The pet starts clueless, wandering around, but through trial-and-error it learns which moves bring rewards and which don’t. By the end, you’ll have a smart companion that finds the fastest route to snacks!

Problem Statement

Help beginners understand the core idea of RL — “learn by exploring and getting feedback”, in a way that’s visual and entertaining, rather than abstract math.

Expected Outcomes

A simple environment (grid or tiny maze) drawn with text or 2D graphics; A reinforcement learning agent; Visualization of: Current policy (best move for each square) and Training progress (reward curves); Optional extras: moving traps, multiple treat types, or “levels.”

Intended Users

Students or hobbyists who want to see an AI agent learn in real time while keeping the codebase small.

Skills You Should Know

- Basic Python
- Understanding of coordinates on a grid

Skills You'll Learn

- RL fundamentals: state, action, reward, exploration vs. exploitation
- Implementing a Q-table and updating it with rewards
- Debugging/tracking agent progress (reward plots)
- Optional: tuning hyperparameters (learning rate, epsilon)

Client / Project Point of Contact

- Tech Lead: Not Provided (students must self-organize)
- Client Point of Contact: Prototyp3 Team, prototyp3.org@gmail.com

Resources & Learning:

What You're Building

- You're teaching a little pet to find treats in a tiny grid world.
- At first, it doesn't know anything, it just wanders.
- Over time, it learns by trying things and getting feedback ("good job" for treats, "oops" for traps).
- By the end, you'll see your pet choose the fastest snack route!

Useful Resources:

- Reinforcement Learning basics:
https://spinningup.openai.com/en/latest/spinningup/rl_intro.html
- Q-Learning explained simply:
<https://www.learndatasci.com/tutorials/reinforcement-q-learning-scratch-python-openai-gym/>
- OpenAI's Gym's FrozenLake:
https://www.gymnasium.dev/environments/toy_text/frozen_lake/

- Gentle Intro to Reinforcement Learning:
<https://www.learndatasci.com/tutorials/reinforcement-q-learning-scratch-python-openai-gym/>
- Short video: Q-Learning explained simply:
<https://www.youtube.com/watch?v=qhRNvCVVJaA>

Tools & Libraries (feel free to use different tech stack or additional ad-ons)

- Python 3.10+ (recommended: use Anaconda)
- NumPy – for math & arrays
- Matplotlib – to plot learning curves
- Optional graphics:
 - Pygame (simple 2D visualization)
 - Tkinter Canvas (comes with Python)

Step-By-Step Tips

1. Set up a folder for your project (or make a GitHub repo).
2. Start very small: print the grid in your terminal using letters or emojis (🐾 for pet, 🍪 for treat).
3. Make the pet move randomly so you understand coordinates.
4. Add rewards:
 - a. +1 for finding a treat
 - b. -1 for stepping on a trap
5. Create a table (think like an Excel sheet) to remember “if I’m here and move there, how good is it?” (that’s the Q-table).
6. Each time the pet plays, update the table to make future choices smarter.
7. Plot your pet’s “score over time” to see it improve.
8. When basics work, add fun extras: more treats, moving traps, or a bigger maze.

Team Work Guide

How to make your 10 weeks together awesome:

1. Set up an introduction call for team bonding.
2. Agree on shared tools for communication and project management.
3. In absence of tech lead, pick a team lead or rotating coordinator to keep things moving.
4. Share frequent updates within the team chat, even if it is a small win or a bug. This builds trust and accountability.

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5. Break your project into small and clear tasks. Assign tasks fairly such that everyone gets a mix of comfort-zone and growth-tasks.
 6. Highly recommended: Have a weekly sync (stand-up).
 7. Use version control (Github) and commit with clear details.
 8. Document everything.
 9. Be respectful to everyone.
 10. HAVE FUN!

Timeline & Milestones:

Item	Start Date	End Date / Deadline
Cohort Period	19th September, 2025	30th November, 2025
First Team/Client Meeting	Week 1	26th September, 2025
Mid-Cohort Update	Week 5	31st October, 2025
Final Submission	Week 9	20th November
Public Project Showcase	Week 10 (Nov 21-28)	TBD
End-of-Cohort Feedback/Interview	December, 2025	TBD

– *Happy Prototyping!*